

— INDUSTRIAL PREDICTIVE QUALITY

From plant data to prioritized decisions

Visual defect detection, machine-risk scoring and process-deviation alerts built from data the plant already produces.

Industrial vision

Sensors & SCADA

Self-supervised learning

Local deployment



Visual inspection

Heatmaps and part ranking



Machine risk

Physical signals and context



Process deviation

Expected vs observed
evolution



Prioritization

Patch → part → cycle → lot →
line



— THE PROBLEM

Signals appear before the cost

The goal is not to generate more data or more alarms. It is to identify what changed, where it changed and what deserves attention first.

01 · SIGNALS

Cameras, vibration, current, SCADA and test logs

Rich but fragmented information at different spatial and temporal scales.



02 · HIDDEN COST

Late defects, scrap, micro-stops and reactive maintenance

Impact becomes visible when the intervention window is already small.



03 · DECISION

A prioritized, explainable and measurable queue

What to inspect, which asset to review and when the deviation started.

Three modules, one decision system

01 • VISION



Visual quality

Learns normal production appearance and localizes deviations on each part.

- Defect heatmaps
- Inspection ranking
- Surface, PCB, packaging or product

02 • SENSORS



Machine risk

Combines physical signals, process variables and calibrated models.

- Top assets/cycles
- Lead time and false alarms
- Engineered + learned features

03 • PROCESS



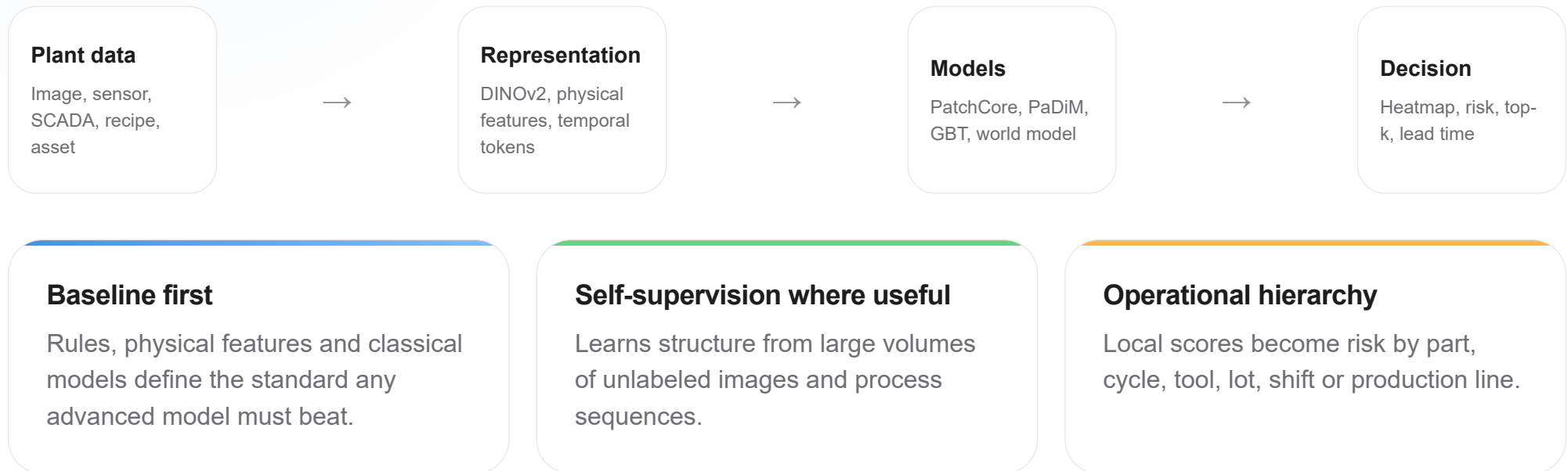
Temporal deviation

Compares expected process evolution with what is actually observed.

- Local surprise and persistence
- Window- or cycle-level change
- Lot- and line-level aggregation

From raw data to an actionable alert

The system combines strong industrial baselines with self-supervised representation learning. Complexity is kept only when it adds measurable value.



Visual inspection that can start with normal images



Dense features + normality memory

DINOv2/ResNet extract local descriptors. PatchCore and PaDiM compare each region with normal history to localize deviations.

- ✓ Image- or part-level score
- ✓ Patch-level heatmap
- ✓ Threshold calibrated on normal production
- ✓ Ranking to reduce manual inspection

Visual inspection queue



Part #A-1842	0.94
Part #A-1837	0.81
Part #A-1819	0.62
Part #A-1811	0.38

Pilot output: ranking, heatmap and comparison against normal references.

A learned signal that **complements** process engineering

The practical standard remains operational context + physical features. The token-level model is retained when it delivers a measurable incremental signal.

0.6609

Metadata-only

Contextual AUPRC baseline in the current diagnostic.

0.6766

Metadata + EWMA surprise

Temporal surprise adds a small but positive signal.

0.6960

Metadata + token world model

Best no-action combination; Δ AUPRC +0.0352.

Learn normal operation without waiting for defects

0

Defect labels required to begin learning normality

For visual representation, anomaly detection and temporal deviation.

1

Unlabeled data already has value

Normal images, sensor sequences and production cycles can teach structure and expected evolution.

2

Labels improve measurement; they are not a universal prerequisite

Defects, scrap and failure events allow recall, lead-time and calibration analysis when available.

3

Two valid outputs

Without labels: anomaly and deviation. With historical events: supervised risk and economic validation.

— TECHNICAL EVIDENCE

Results that justify validation on real plant data

1.00

Visual AUPRC

DINOv2 + PatchCore/PaDiM in a quick MVTec bottle run.

+0.0352

Sensor delta

Token world model over metadata-only in the current diagnostic.

4–6

Weeks

To audit, benchmark, model and deliver a go/no-go recommendation.

What this supports

The visual architecture produces a strong public-benchmark signal, and the dense sensor model adds incremental temporal information.

What we do not assume

A public result does not guarantee performance on another line. Value is confirmed on the client's real distribution, defects and operating conditions.

— PILOT

A short path to an investment decision

WEEK 1

Data map

Inventory, quality, traceability, temporal structure and use-case definition.

WEEKS 1-2

Baselines

Rules, classical models and a strong visual foundation to set a realistic reference.

WEEKS 2-5

Modeling

Heatmaps, risk scoring, temporal surprise, calibration and error analysis.

WEEK 6

Decision

Local demo, final benchmark, expected impact and integration roadmap.

— DELIVERABLES

A demonstration the team can see, measure and challenge

- ✓ Data manifest and quality audit.
- ✓ Benchmark against rules and strong baselines.
- ✓ Visual heatmaps and inspection ranking.
- ✓ Asset/cycle risk and temporal deviation curves.
- ✓ Operational metrics: precision@top-k, false alarms and lead time.
- ✓ Local demo, executive report and technical documentation.
- ✓ Go/no-go recommendation and integration architecture.

Top operational alerts

local demo

Line 2 · Lot 184		High
Machine 04 · Cycle 912		High
Part B-3381		Review
Night shift · Process 7		Watch

— STARTING DATA

Start with what exists; labels are optional

MINIMUM TO BEGIN

- Normal images or process time series.
- Timestamps, part/asset/cycle identifiers and temporal ordering.
- Available operating context: recipe, material, setpoints or machine.
- A technical owner who can interpret the process and its limits.

INCREASES VALUE, BUT DOES NOT BLOCK THE PILOT

- Labeled defects or inspection outcomes.
- Failure, scrap, rework or maintenance history.
- Cost of false alarms and missed defects.
- Verified temporal commands for action-conditioned evaluation.

Self-supervision learns normality without labels. Historical events are used when a specific supervised target must be measured or calibrated.

— SUCCESS CRITERIA

The pilot ends with a quantitative answer



Less unnecessary inspection

Focus manual review on the parts or cycles with the highest measured risk.



Earlier warning

Detect deviation before the defect, stoppage or reactive maintenance event.



Defensible go / no-go

Decide whether to integrate, collect more data or stop without relying on black-box promises.

An industrial validation platform: strong baselines, AI where it adds value, and traceable results.